

Seeing Data: Visualisation, Intelligence, and Interpretation

Core question

How do charts create meaning?

Core message

A chart is a claim about how the world is organised. The claim is produced through selection, encoding, comparison, framing and omission.

Learning outcomes

By the end of this session students should be able to:

1. Explain how selection, encoding and framing turn data into meaning.
2. Distinguish data, encoding, observation, insight and narrative, and a fluent interpretation from a justified insight.
3. Explain why identical summary statistics can conceal different structures.
4. Separate a visible pattern from an unsupported causal claim.
5. Use an AI assistant to produce a chart and verify every claim it makes about the data.
6. Critique a human-made chart and an AI-generated interpretation using course terminology.
7. Record significant intelligent-tool use with the course disclosure block.

Preparation

Students need a Python notebook environment with `pandas`, `matplotlib` and `numpy`. An AI assistant is expected but every task remains possible without one.

Instructor materials:

- the Seeing Data deck (`beamer/seeing-data.pdf`; rebuild with `lualatex seeing-data.tex` in `beamer/`) and `slides.md` speaker directions;
- the lab notebook opened and run once before class;
- the Critique and Repair activity sheet and critique pack (`activities/`);
- the chart card pack (`activities/chart_cards.pdf`) printed single-sided, each pair draws three cards in the studio;
- originals embedded in the deck: Snow 1854 (public domain) and the Hawkins warming stripes (CC BY-SA 4.0); every other figure is a course reproduction generated from data.

Three-hour run sheet (matches the deck order)

Time	Segment	Deck slides	Teaching purpose
0:00–0:08	Private Canvas preference survey + subject workflow	3–4	Complete the private preference task; state permitted AI workflow and boundaries
0:08–0:15	Premise and outcomes	5–7	A chart is a claim; five decisions produce it

Time	Segment	Deck slides	Teaching purpose
0:15–0:30	Anscombe prediction and reveal	8–9	Establish that summaries are selective
0:30–0:38	Vocabulary: data → encoding → observation → insight → narrative	10	Establish shared vocabulary
0:38–0:50	Claim sprint: twelve charts, 40 seconds each	11–23	Every chart is a claim; students write two “the chart shows...” sentences; debrief two
0:50–1:05	Snow worked example + the shows/suggests debrief	24–26	
1:05–1:13	Six ways a true chart misleads	27–28	Separate spatial pattern, inference and causal evidence
1:13–1:22	AI interpretation and the verification ladder	29–32	Accurate charts can still deceive; the window is part of the claim
1:22–1:30	Think · pair · share	33	Fluency does not guarantee validity
1:30–1:40	Break	34	A found chart against the six mechanisms
1:40–2:30	Notebook lab (<code>notebooks/seeing_data_lab.ipynb</code>)	35	, Calculate, plot, generate, verify, repair
2:30–2:50	Studio: chart cards + Critique and Repair	36	
2:50–2:55	Pair exchange (Critique 1 frame)	37	Three cards per pair: claim, audience, what could mislead; then the repair artefact
2:55–3:00	Close the loop + exit ticket	38	Test whether claims are supported and specific
			Record a decision, a risk and a verification need

The diagnostic quiz runs on Canvas after class, open book, closed assistant, ungraded; it calibrates the cohort, not marks.

Content topics

- visualisation as selection and interpretation
- human and machine intelligence in visualisation
- generated charts and hallucinated insight
- the five-step verification ladder for generated output

Facilitation notes

- Do not show Anscombe’s plots before students predict their appearance.

- Ask students to use “the chart shows” only for visible evidence. Reserve “this suggests” for interpretation.
- Avoid presenting Snow’s map as proof by itself. Ask what additional evidence supports causation.
- During assistant use, require students to retain their initial prediction and prompt summary.

Notebook

Three tiers in notebooks/: distribute `seeing_data_lab.ipynb` at the start of the lab; hold `seeing_data_starter.ipynb` as an intervention for students blocked by syntax; keep `seeing_data_instructor.ipynb` until the activity and any related assessment have concluded. Core exercises:

1. Summary statistics vs the picture (Anscombe).
2. Assistant-generated chart of the transport dataset plus verification checks.
3. Claim audit of a generated interpretation, four-way: supported directly / plausible but unverified / unsupported / contradicted.

Bertin’s visual variables are introduced only as a bridge here; channel precision and visual-task selection are taught in depth in Choosing Visual Forms.

Studio

Primary: **Critique and Repair** (`activities/activity_critique_and_repair.md` with `activities/critique_pack.md`). Warm-up: each pair draws three cards from `activities/chart_cards.pdf` and answers, per card: the claim, the audience, one way it could mislead. Then the full read–test–verify–repair cycle on the satisfaction artefact, with the responsibility checkpoint.

Pair exchange uses the **Critique 1** frame (`activities/critique_1_human_vs_ai.md`): one human-made chart vs one AI-generated interpretation. This is the direct rehearsal for A1 parts A and C.

Evidence of learning

Collect: completed notebook checks; one repaired caption; the Critique and Repair sheet; the exit ticket.

Exit ticket

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One design decision I made today:
One risk or limitation I noticed:
One thing I need to verify before next class:
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References

Required: Cairo, *How Charts Lie* (selected chapter); D’Ignazio & Klein, *Data Feminism* (selected chapter).

Recommended: Munzner, “A Nested Model for Visualization Design and Validation”; Hicks, Humphries & Slater, “ChatGPT is bullshit”; Bender et al., “On the Dangers of Stochastic Parrots”.